

DBDiagram

```
1 // Use DBML to define your database structure
2 // Docs: https://dbml.dbdiagram.io/docs
3
4 Table libros {
5   id integer [primary key]
6   titulo varchar [not null]
7   autor varchar [not null]
8 }
9
10 Table prestamos {
11   id integer [primary key]
12   fecha_prestamo timestamp [not null]
13   id_usuario integer [not null]
14   id_libro integer [not null]
15 }
16
17 Table usuarios {
18   id integer [primary key]
19   nombre varchar [not null]
20   apellido varchar [not null]
21   email varchar [not null]
22 }
23
24 Ref: usuarios.id < prestamos.id_usuario
25 Ref: libros.id < prestamos.id_libro
26
27 |
```



Creación de contenedor

```
biblioteca_municipal
PS C:\Users\angel> docker run --name biblioteca_municipal -e MYSQL_ROOT_PASSWORD=root -p 3307:3306 mysql
2026-08-29 02:19:47+00:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 26.7.0-1.el9 started.
```

MySQL

```
PS C:\Users\angel> docker exec -it biblioteca_municipal bash
bash-5.1# mysql -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 9
Server version: 26.7.0 MySQL Community Server - GPL

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> show databases;
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
| sys |
+-----+
4 rows in set (0.007 sec)

mysql> use biblioteca
ERROR 1049 (42000): Unknown database 'biblioteca'
mysql> use biblioteca;
ERROR 1049 (42000): Unknown database 'biblioteca'
mysql> create database biblioteca;
Query OK, 1 row affected (0.012 sec)

mysql> use biblioteca;
Database changed
mysql> CREATE TABLE libros (id INTEGER PRIMARY KEY, titulo VARCHAR NOT NULL, autor VARCHAR NOT NULL);
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'N
OT NULL, autor VARCHAR NOT NULL)' at line 1
mysql> CREATE TABLE libros (id INT PRIMARY KEY, titulo VARCHAR(255) NOT NULL, autor VARCHAR(255) NOT NULL);
Query OK, 0 rows affected (0.052 sec)
```

```

mysql> CREATE TABLE libros (id INTEGER PRIMARY KEY, titulo VARCHAR NOT NULL, autor VARCHAR NOT NULL);
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'NOT NULL, autor VARCHAR NOT NULL)' at line 1
mysql> CREATE TABLE libros (id INT PRIMARY KEY, titulo VARCHAR(255) NOT NULL, autor VARCHAR(255) NOT NULL);
Query OK, 0 rows affected (0.052 sec)

mysql> CREATE TABLE usuarios (id INT PRIMARY KEY, nombre VARCHAR(255) NOT NULL, apellido VARCHAR(255) NOT NULL, email VARCHAR(255) NOT NULL);
Query OK, 0 rows affected (0.037 sec)

mysql> CREATE TABLE prestamos (id INT PRIMARY KEY, fecha_prestamo TIMESTAMP NOT NULL, id_usuario INT NOT NULL, id_libro INT NOT NULL, FOREIGN KEY (id_usuario) REFERENCES usuarios(id), FOREIGN KEY (id_libro) REFERENCES libros(id));
Query OK, 0 rows affected (0.036 sec)

mysql> INSERT INTO libros (id, titulo, autor) VALUES (1, 'Cien aos de soledad', 'Gabriel Garca Mrquez'), (2, '1984', 'George Orwell'), (3, 'El principito', 'Antoine de Saint-Exupry'), (4, 'Don Quijote de la Mancha', 'Miguel de Cervantes');
Query OK, 4 rows affected (0.037 sec)
Records: 4 Duplicates: 0 Warnings: 0

mysql> INSERT INTO usuarios (id, nombre, apellido, email) VALUES (1, 'Juan', 'Prez', 'juan.perez@email.com'), (2, 'Mara', 'Gmez', 'maria.gomez@email.com'), (3, 'Carlos', 'Lpez', 'carlos.lopez@email.com'), (4, 'Ana', 'Martnez', 'ana.martinez@email.com');
Query OK, 4 rows affected (0.016 sec)
Records: 4 Duplicates: 0 Warnings: 0

mysql> INSERT INTO prestamos (id, fecha_prestamo, id_usuario, id_libro) VALUES (1, '2026-08-20 10:30:00', 1, 2), (2, '2026-08-21 14:15:00', 2, 1), (3, '2026-08-22 09:00:00', 3, 4), (4, '2026-08-23 16:45:00', 1, 3), (5, '2026-08-25 11:20:00', 4, 2);
Query OK, 5 rows affected (0.017 sec)
Records: 5 Duplicates: 0 Warnings: 0

mysql> show tables;
+-----+
| Tables_in_biblioteca |
+-----+
| libros                |
| prestamos             |
| usuarios              |
+-----+
3 rows in set (0.005 sec)

```

```
mysql> SELECT * FROM libros;
```

id	titulo	autor
1	Cien aos de soledad	Gabriel Garca Mrquez
2	1984	George Orwell
3	El principito	Antoine de Saint-Exupry
4	Don Quijote de la Mancha	Miguel de Cervantes

```
4 rows in set (0.001 sec)
```

```
mysql> SELECT * FROM prestamos;
```

id	fecha_prestamo	id_usuario	id_libro
1	2026-08-20 10:30:00	1	2
2	2026-08-21 14:15:00	2	1
3	2026-08-22 09:00:00	3	4
4	2026-08-23 16:45:00	1	3
5	2026-08-25 11:20:00	4	2

```
5 rows in set (0.010 sec)
```

```
mysql> SELECT * FROM usuarios;
```

id	nombre	apellido	email
1	Juan	Prez	juan.perez@email.com
2	Mara	Gmez	maria.gomez@email.com
3	Carlos	Lpez	carlos.lopez@email.com
4	Ana	Martnez	ana.martinez@email.com

```
4 rows in set (0.001 sec)
```